ZISHAN WANG

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Career Interests: Clinical Trails, Epidemiology, Statistical Modeling

SUMMARY OF QUALIFICATIONS

M.S. student in Biostatistics at the University of Washington with a B.S. background in Biotechnology and interdisciplinary experience in human disease research.
Research experience in quantitative analysis of disease-related datasets, including bioinformatics workflows, statistical analysis, and data visualization using R.
Experience interpreting biological and health data through human disease research, with an interest in applying computational methods to improve population health outcomes.
Ability to communicate and collaborate with interdisciplinary research teams.

EDUCATION

University of Washington(MS) WA,US
M.S. in Biostatistics, Data science track 09/2026-05/2028

- Relevant Coursework: Probability theory; Statistical Inference; R; SQL(Structured Query Language); SAS(Statistical Analysis System)

Southwest University (SWU) (BS) Chongqing, CHN
B.S. in Biotechnology (Joint Program with University of Western Australia (UWA)) 09/2022-06/2026
Grade: 86.43/100 | Major Rank:5/48

- Relevant Coursework: Biostatistics, Advanced Calculus; Biochemistry, Molecular Biology, Cell Biology and Immunology; UC San Diego Division of Extended Studies: Calculus II and Linear Algebra

PROFESSIONAL EXPERIENCE

Bioinformatics Analysis of Pancreatic Cancer 04/2025-09/2025
Research Assistant, First Author; Supervised by Professor Guangwei Zhang, Department of Neuroscience, University of California

- Analyzed and interpreted large-scale health datasets from peripheral blood samples of healthy individuals and pancreatic ductal adenocarcinoma (PDAC) patients
- Performed data quality control, harmony integration to eliminate batch difference, cell type annotation, differential gene analysis, and biological process enrichment through R and online platforms like Shiny Go
- Integrated and interpreted analytical results from previous studies to explore associations between disease patterns and peripheral blood cell indicator. Drafted conference papers, published and created academic posters, and delivered on-site presentations

2024 iGEM: Use Deep Learning to Guide the Research of Antifungal Polypeptides 12/2023-10/2024
Wet Lab Group Leader; Joint Project---State Key Laboratory of Resource Insects, Biological Science Research Center, Academy for Advanced Interdisciplinary Studies, SWU, and College of Pharmaceutical Sciences, SWU

- Built a DeBERTa-based deep learning pipeline for high-throughput screening of antifungal peptides against Candida albicans to address fungal infections and antifungal resistance
- Performed MIC assays and murine skin infection models to evaluate efficacy and toxicity of AI-predicted peptides. Assessed antifungal activity, cytotoxicity, and therapeutic efficacy of antimicrobial peptides in a mouse of skin disease
- Conducted statistical analysis and visualization of experimental data using GraphPad and BioRender, and created science communication materials and illustrations for the general public and non-professional audiences

ChiaTai Tianqing Pharmaceuticals 03/2026-06/2026
Clinical Research Associate (CRA)

- Developed a solid understanding of clinical trial workflows, Good Clinical Practice(GCP) guidelines, and SOPs related to clinical research operations
- Independently managed and monitored two Phase II clinical trials for allergic rhinitis therapies, participating in the full study process from Study Start-Up(SSU) initiation to subject close-out; the investigational product has now progressed into Phase III trials
- Collaborated with Clinical research coordinators(CRC), project managers, hospital department directors, and clinical site teams, gaining hands-on experience in clinical research operations and cross-functional communication

Composition of Bacteria and Identification of Key Strains in RC1 Microalgal-Bacterial Consortium
Leader; Sericulture Laboratory, College of Sericulture, Textile and Biomass Sciences, SWU 01/2024-01/2026

- Analyzed microbial community structure in the microalgae-bacterial consortium, using metagenomic techniques to evaluate community diversity and its correlation with nitrogen and phosphorus removal efficiency
- Integrated existing research findings to perform large-scale statistical analyses, including environmental factor assessment and correlation analysis, to investigate associations between environmental determinants and biological outcomes

ADDITIONAL QUALIFICATIONS

Skills

- SPSS, R, GraphPad Prism, SQL, SAS, Python
- Bioinformatics: Single-cell RNA sequencing analysis, GEO database analysis, Harmony-based batch correction, ShinyGO functional enrichment analysis, and SnapGene sequence analysis.
- Fluent in both Mandarin Chinese and English

Honors

- SWU Excellence Award, China International College Students' Innovation and Entrepreneurship Competition (06/2026)
- SWU Third Prize, China International College Students' Innovation Competition (08/2024)
- SWU Second Prize, China National Undergraduate "Innovation, Creativity and Entrepreneurship" Challenge (05/2024)
- SWU Third-class Scholarship (11/2024 and 11/2025)

Publications

Zishan Wang. Single-Cell Sequencing Reveals the Interplay Between Microbiota, Zinc Metabolism, and Immune Dysregulation in Pancreatic Ductal Adenocarcinoma. *2026 International Symposium on Biological Neural Networks and Intelligent Optimization.* Published.